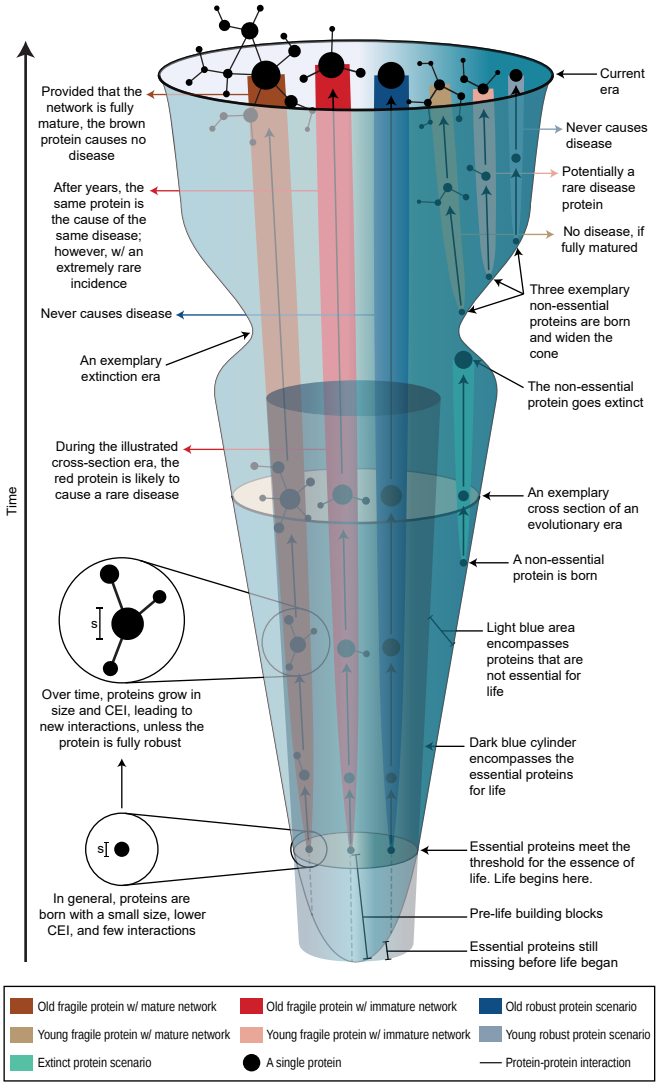

Time
Current era
Provided that the network is fully mature, the brown protein causes no disease
Never causes disease
After years, the same protein is the cause of the same disease; however, w/ an extremely rare incidence
Potentially a rare disease protein
No disease, if fully matured
Three exemplary non-essential proteins are born and widen the cone
Never causes disease
An exemplary extinction era
The non-essential protein goes extinct
During the illustrated cross-section era, the red protein is likely to cause a rare disease
An exemplary cross section of an evolutionary era
A non-essential protein is born
s
Light blue area encompasses proteins that are not essential for life
Over time, proteins grow in size and CEI, leading to new interactions, unless the protein is fully robust
Dark blue cylinder encompasses the essential proteins for life
s
Essential proteins meet the threshold for the essence of life. Life begins here.
In general, proteins are born with a small size, lower CEI, and few interactions
Pre-life building blocks
Essential proteins still missing before life began
Old fragile protein w/ mature network
Old fragile protein w/ immature network
Old robust protein scenario
Young fragile protein w/ mature network
Young fragile protein w/ immature network
Young robust protein scenario
Extinct protein scenario
A single protein
Protein-protein interaction